

Andrew Freeman

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[Project Portfolio](#)

Mechanical Engineering student (Concordia, 2027) with 3+ years designing and validating high-voltage battery systems, thermal models, and electromechanical hardware for competitive Formula SAE electric vehicles. Built a 369.6 V pack from cell selection through ANSYS-validated thermal models, then raced it. Fluent in Python, MATLAB, SolidWorks, and hands-on lab instrumentation.

TECHNICAL SKILLS

Hardware & Test: Electronic loads • Oscilloscopes • DCIR/HPPC/capacity testing • Thermal instrumentation • Tensile test fixtures (ASTM)

Simulation: ANSYS Fluent (CHT, airflow) • RC electrical models • MATLAB (pump curves, thermal, electrical)

CAD: SolidWorks • Fusion 360 • AutoCAD (DFM)

Programming: Python (automation, data logging, control scripts) • MATLAB • C, C++ (embedded systems)

Systems: Full-vehicle electromechanical integration (HV battery, BMS, cooling, chassis, suspension, drivetrain, LV electrical, vehicle controls) • Sensor integration (analog/digital) • GDB firmware debugging

OS: Windows • Linux

Languages: English — fluent • French — fluent • Spanish — fluent

EDUCATION

B.Eng. Mechanical Engineering · *Concordia University*

2022 – 2027

Relevant courses: Heat Transfer · Machine Element Design · Microcontrollers · Properties & Failures of Material · Applied Differential Equations · Applied Statistics

EXPERIENCE

Technical Director/High Voltage Lead · *Concordia Formula Racing*

Jun 2024 – Present

Led all engineering sub-teams (chassis, drivetrain, suspension, HV/LV electrical) for an electric FSAE vehicle.

- ▶ Designed a 369.6 V lithium-ion battery pack end-to-end: cell selection, enclosure, BMS integration and validation (replacing a prior design)
- ▶ Cut cell-characterization timeline by 40% by authoring Python scripts that automated DCIR/capacity test cycles on electronic loads, eliminating manual data transcription.
- ▶ Built a coupled RC-electrical + conjugate heat-transfer thermal model (MATLAB + ANSYS Fluent) that predicted pack temperature within ± 3 °C across endurance-simulation runs, reducing physical test iterations.
- ▶ Modelled regen braking and implemented real time calculation and controls on custom controller in embedded C.
- ▶ Validated structural enclosure materials at operating temperatures: designed furnace-heated tensile test fixtures per ASTM standards, confirming yield-strength and Young's modulus assumptions used in CAD safety margins.
- ▶ Identified and corrected a critical electrical fault in vehicle control boards via GDB-assisted firmware debugging, preventing DNF at competition.
- ▶ Reviewed all mechanical subsystem designs (chassis, suspension, drivetrain) against system-level integration requirements, preventing over-engineered components by enforcing appropriate FOS targets and re-running FEA with real-world boundary conditions including bolt preload, catching discrepancies before manufacturing.
- ▶ Redesigned and personally welded the chassis to house a battery pack 4" smaller in both length and width, reducing wheelbase from 1,575 mm to 1,540 mm (tighter turning radius) and cutting chassis mass by 500 g while satisfying all FSAE structural and packaging regulations.

Cooling System Lead · *Concordia Formula Racing**Jun 2023 – Jun 2024**Owned thermal design for the team's first competitive EV, with zero prior baseline to work from.*

- ▶ Delivered a cooling system over 50% lighter than the previous combustion-era design by building a MATLAB pump-selection model that optimized head curves, pressure losses, and duty cycle simultaneously.
- ▶ Designed and integrated a real-time DAQ system using thermistors and Hall-effect flow sensors, enabling instantaneous kW heat-rejection measurement - data that directly informed the 2024 pack thermal model.
- ▶ Optimized pump duty cycle through the MATLAB model, reducing average cooling-system current draw and extending projected battery range under endurance-event conditions.

Lab Intern · *Nitrex Metal Inc.**May 2023 – Jan 2025*

- ▶ Led a salt-bath corrosion-testing project evaluating seven heat-treatment process variants; statistical analysis of mechanical-property variance provided actionable process-optimization recommendations adopted by the QA team.
- ▶ Maintained traceability documentation for a commercial heat-treatment facility operating under strict quality-compliance protocols, achieving zero audit findings across tenure.

Pit Crew Technician · *SH-Karting**Jul 2024 – Present*

- ▶ Maintained mechanical reliability of Porsche 944 and 911 race vehicles under continuous hot lap conditions; rapid trackside diagnostics including brake service and EFI/carb fuel-mixture correction, keeping vehicle downtime near zero.

PROJECTS

Amazon DeepRacer — Reinforcement Learning ·**2023**

- ▶ Trained an RL driving model on a self-hosted Linux server, combining vehicle-dynamics constraints with reward-function tuning; finished 67th out of 2,325 entrants (top 3%) in the unranked division.